

Serena Noboudem

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EDUCATION

Boston University, *Masters in Computer Science, Cybersecurity Specialization*

Sep 2025 – May 2027 | Boston, MA

- **Awards:** Merit-based Scholarship Recipient
- **Coursework:** CS630: Graduate Algorithms, CS655: Computer Networks, CS611: Software Principles & Design

Kennesaw State University | **GPA - 3.79**,

May 2025 | Kennesaw, GA

Bachelor of Science in Computer Science

SKILLS

- **Programming & Frameworks:** Java, Spring Boot, Python, Kotlin, HTML5/CSS3, JavaScript/TypeScript, React
- **DevOps & Tools:** Git/GitHub, CI/CD (GitHub Actions, Jenkins), IntelliJ IDEA, VSCode, Kubernetes, Docker
- **Backend, Database & API Dev:** MySQL, PostgreSQL, Firebase, AWS S3, Redis, RESTful API, Webclient
- **Security & Networking:** Secure coding, Threat modeling, Authentication/Authorization (OAuth2, RBAC, JWT), Network protocols (DNS, DHCP, TCP/IP), Wireshark

WORK EXPERIENCE

Undergraduate Research Assistant, *KSU - College of Computing*

Oct 2024 – May 2025

Abstract 📄 | *Poster* 📄

- Crafted & executed adversarial attacks (MIFGSM, FGSM) on deep learning models using the MNIST dataset, executing **40+ attack iterations**, applying **threat modeling techniques** to assess model robustness and uncover key vulnerabilities
- Conducted **cross-model evaluation** across 4+ architectures, analyzing **5,000+ misclassified samples**, visualizing error patterns, and improved **model robustness by 80%**
- Presented research findings at the **Annual KSU Symposium**, effectively translating complex technical concepts into accessible insights for an audience of **50+** faculty, researchers, and peers.

Software Engineer Intern - Security, *Afriland First Bank*

Jun 2023 – Aug 2023

- Performed **threat modeling** using the **STRIDE methodology** to identify and prioritize system vulnerabilities, resulting in the mitigation of **15+** high-priority security risks before deployment.
- Engineered a **secure authentication system** using **OAuth2, JWT**, and **2FA** in Java for a banking app, achieving **100% compliance** with banking regulations and reducing unauthorized access attempts by **70%**.
- Integrated automated security testing using **SonarQube** into CI/CD pipelines for static code quality analysis, reducing critical code vulnerabilities and bugs by **60%** pre-deployment.

Software Engineer, Intern,

Jun 2022 – Oct 2022

PKF Technology Center (IT & Development Company)

- Built and maintained a secure **Spring Boot RESTFUL API** for an LMS, handling **100+ requests/seconds** and improving data retrieval speeds by **25%**
- Designed and delivered an interactive web-based analytical dashboard using **HTML, CSS**, and **JavaScript**, with data visualization features, improving user satisfaction by **30%** through enhanced data-driven decision-making capabilities
- Drove engineering best practices through active code reviews, secure coding guidelines, and automated static code analysis, resulting in a **25%** reduction in bug reports and improved code maintainability

PROJECTS

Mobile D2L (Kotlin | Firebase)

- Built a modular Android app simulating the D2L course system using Kotlin, incorporating **13+ content types**, improving navigation efficiency by **40%** for **50+** students.
- Integrated **Firebase Authentication** with secure login flows and Firebase Database, ensuring data integrity and reducing backend development time by **60%**

FinHub (Spring boot | React.js | MySQL)

- Developed a full-stack finance tracker web app in **Spring Boot** and **React.js** to retrieve and analyze financial data from **2+** bank accounts, to generate personalized spending insights and budget reports
- Integrated the Plaid API via **WebClient** with secure authentication, enhancing HTTP request handling efficiency by **10%** and ensuring safe, scalable access to user data with reduced latency

DNS Server (Python)

- Engineered a multithreaded DNS server in Python using **Socket programming**, handling **500+ QPS**, achieving **99%** resolution accuracy through efficient request parsing and high-throughput
- Implemented multi-stage request validation to detect and block malformed queries, **mitigating 90% attack** attempts and reducing response latency by **40%**.

CERTIFICATES

- ISC2: Certified In Cybersecurity - In Progress, Expected November 2025 📄